



$$U_N = \frac{2^{\frac{1}{N}}}{N(2^{\frac{1}{N}} - 1)}. \quad [4]$$

[illegible]

- (b) Use a similar method to find, in terms of N , a lower bound L_N for $\int_0^1 2^x dx$. [4]

[illegible]

- (c) Find the least value of N such that $U_N - L_N < 10^{-4}$. [2]

[illegible]